

## Drill Problems 14

*Author: Jaehoon Song**Release: 2025-08-10 15:12:04-04:00***Purpose and Usage:**

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**C O L U M B I A   A C A D E M Y***enrichment beyond the classroom*

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Written by Jaehoon Song

1. **Absolute Value Equation** (10 points)

Consider the equation:

$$4|12 - 2x| + 12|6 - x| = 140$$

What is the positive solution to the given equation?

**Answer:**



2. **Data Set Comparison** (10 points)



Figure 1: reference attached

The histogram shows the data collected from set  $X$ . Data set  $Y$  takes all the values from set  $X$  and subtracts 17.5 to every value. Which of the following correctly compares the median and the range between set  $X$  and set  $Y$ ?

- (A) The median for set  $X$  is greater than the median for set  $Y$  and the range for set  $X$  is greater than the range of set  $Y$ .
- (B) The median for set  $X$  is greater than the median for set  $Y$  and the range for set  $X$  is equal to the range of set  $Y$ .
- (C) The median for set  $X$  is equal to the median for set  $Y$  and the range for set  $X$  is greater than the range of set  $Y$ .
- (D) The median for set  $X$  is equal to the median for set  $Y$  and the range for set  $X$  is equal to the range of set  $Y$ .

**Answer:**



3. **Exponential Function Transformation** (10 points)

Function  $g$  is defined by the equation  $g(x) = 8\left(\frac{1}{4}\right)^x$ . If  $h(x) = g(x+2)$ , which of the following is the equation for  $h(x)$ ?

- (A)  $h(x) = \frac{1}{2}\left(\frac{1}{4}\right)^x$
- (B)  $h(x) = 4\left(\frac{1}{4}\right)^x$
- (C)  $h(x) = 8\left(\frac{1}{4}\right)^x + 2$
- (D)  $h(x) = 16\left(\frac{1}{2}\right)^x$

**Answer:**

□

4. **Linear Function Problem** (10 points)

The linear function  $f(x)$  is defined such that  $f(2) = 13.5$  and  $f(14) = 22.5$ . If  $f(x+1) = 18.75$ , what is the value of  $x$ ?

**Answer:**

□

5. **Linear Equation Solution** (10 points)

Consider the equation:

$$0.5(1.5x - 0.75) = 3\left(-\frac{x}{6} + \frac{1}{12}\right)$$

What is the solution to the given equation?

**Answer:**

□

6. **Quadratic Equation Solution** (10 points)

Consider the equation:

$$x^2 - 12x + 7 = 0$$

One solution to the given equation can be written as  $j - \sqrt{k}$ , where  $j$  and  $k$  are constants. What is the value of  $jk$ ?

**Answer:**

□

7. Function Graph Translation (10 points)

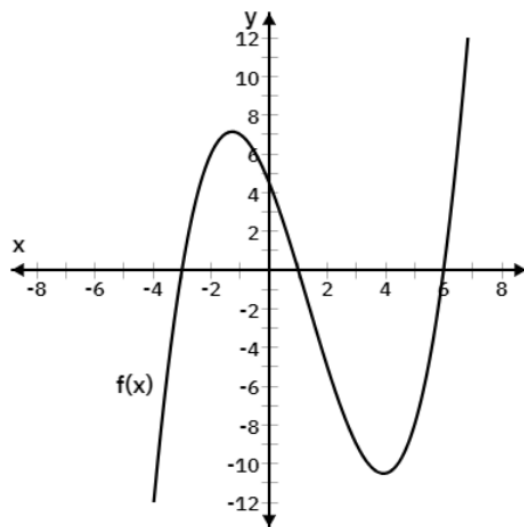


Figure 2: reference attached

The graph of  $y = f(x)$  is graphed on the  $xy$ -plane. The equation of  $f(x)$  is represented by  $f(x) = a(x + m)(x + n)(x + o)$ , where  $a, m, n$ , and  $o$  are constants. If the graph of  $f(x)$  is moved one unit to the left, which of the following is true?

- (I) The value of  $a$  decreases by 1
  - (II) The product of  $m, n$ , and  $o$  is 0
  - (III) The  $y$ -intercept of the translated graph is at the origin
- (A) I only  
(B) I, II, and III  
(C) II only  
(D) II and III

**Answer:**

□

8. **Polynomial Function Analysis** (10 points)

The table above shows several values of the function  $f$  graphed on the  $xy$ -coordinate plane. The equation

| $x$ | $f(x)$ |
|-----|--------|
| -5  | -90    |
| -3  | 0      |
| 0   | 0      |
| 2   | -20    |
| 4   | 0      |
| 5   | 40     |

of  $f$  is defined by  $f(x) = (x + a)(x + b)(x + c)$ , where  $a$ ,  $b$ , and  $c$  are constants. Which of the following is true?

- (I)  $a + b + c < 0$
  - (II)  $f(-2) < 0$
  - (III)  $f(-6) < f(6)$
- (A) I only
  - (B) I and III
  - (C) I, II and III
  - (D) III only

**Answer:**

□

9. **Exponential Growth Problem** (10 points)

Consider the function:

$$f(t) = 155(1.14)^{\frac{4}{3}t}$$

In the given function, the growth in revenue at a certain company is modeled, where  $t$  represents years. According to the model, how many months does it take for the revenue to increase by 14%?

**Answer:**

□

10. **Geometric Angle Problem** (10 points)

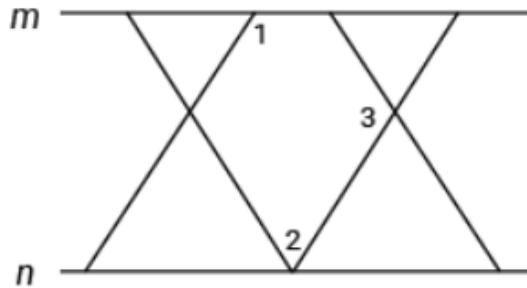


Figure 3: reference attached

In the image shown, line  $m$  is parallel to line  $n$ , all four triangles formed by the intersecting lines are isosceles triangles, and the two smaller triangles are congruent. If the measure of angle 3 is  $104^\circ$ , what is the sum of angle 1 and angle 2, in degrees?

- (A) 128
- (B) 204
- (C) 208
- (D) 256

**Answer:**



11. **System of Inequalities** (10 points)

Let  $t$  represent the number of tulips and  $r$  represent the number of roses in a bouquet. Each tulip costs \$3.50 and each rose costs \$5.25. Flora wants to buy a bouquet for at most \$50 and have at least three times as many tulips as roses in the bouquet. Which system of inequalities best represents this scenario?

- (A)  $3.50t + 5.25r \leq 50$   
 $t \leq 3r$
- (B)  $3.50t + 5.25r \leq 50$   
 $3t \leq r$
- (C)  $3.50t + 5.25r \leq 50$   
 $t \geq 3r$
- (D)  $3.50t + 5.25r \leq 50$   
 $3t \geq r$

**Answer:**



12. **Quadratic Vertex Problem** (10 points)

A quadratic equation with a leading coefficient of 1 intersects the  $x$ -axis at the points  $(a, 0)$  and  $(b, 0)$ . What is the value of the  $y$ -coordinate of the vertex?

- (A)  $-\frac{1}{4}(b - 3a)(b - a)$
- (B)  $-\frac{1}{4}(a - b)^2$
- (C)  $\frac{1}{4}(a - b)^2$
- (D)  $\frac{1}{4}(b - 3a)(b - a)$

**Answer:**

□

13. **Cylinder Surface Area Problem** (10 points)

Two similar cylinders, Cylinder  $A$  and Cylinder  $B$ , have volumes, in cubic meters, of  $100\pi$  and  $800\pi$  respectively. If the radius of Cylinder  $A$  is equal to 2 meters, what is the total surface area of Cylinder  $B$  in square meters? (Surface area of a cylinder =  $2\pi r(r + h)$ )

- (A)  $216\pi$
- (B)  $232\pi$
- (C)  $432\pi$
- (D)  $864\pi$

**Answer:**

□

14. **Absolute Value Function Equation** (10 points)

The function  $f$  is defined by the equation:

$$f(x) = |8 - 5x|$$

For what value of  $k$  does  $f(k) = 6k + 3$ ?

**Answer:**

□

15. **Arcade Group Pricing** (10 points)

For groups of 28 or more people, an arcade charges \$9 per person for the first 28 people and \$15 for each additional person. How many total people were in attendance if the group paid \$447 to go to the arcade?

**Answer:**

□

16. **Circle Inscribed in a Square** (10 points)

The equation

$$3x^2 + 18x + 3y^2 - 6y - 15 = 0$$

gives the graph of a circle in the  $xy$ -plane. If the circle is inscribed in a square, what is the area of the square?

**Answer:**

□

17. **Exponential Function Rewriting** (10 points)

The function  $f$  is defined by the given equation:

$$f(x) = 5(0.92)^{3x}$$

The equation can be rewritten as  $f(x) = 5\left(1 - \frac{p}{100}\right)^x$ , where  $p$  is a constant. Which of the following is closest to the value of  $p$ ?

- (A) 8
- (B) 12
- (C) 22
- (D) 24

**Answer:**



18. **System of Equations with Parameter** (10 points)

Consider the system of equations:

$$\begin{aligned}y &= 3x^2 - 5x - 12 \\ 40x - 4y &= a\end{aligned}$$

In the given system of equations,  $a$  is a positive constant. The system has exactly one distinct real solution. What is the value of  $a$ ?

**Answer:**



19. **Complex Number Simplification** (10 points)

If the expression

$$\frac{8 - i}{3 - 2i}$$

is rewritten in the form  $a + bi$ , where  $a$  and  $b$  are real numbers, what is the value of  $a$ ? (Note:  $i = \sqrt{-1}$ )

**Answer:**



20. **Coordinate Geometry Problem** (10 points)

In the  $xy$ -plane, the point  $(p, r)$  lies on the line with equation  $y = x + b$ , where  $b$  is a constant. The point with coordinates  $(2p, 5r)$  lies on the line with equation  $y = 2x + b$ . If  $p \neq 0$ , what is the value of  $\frac{r}{p}$ ?

- (A)  $\frac{2}{5}$
- (B)  $\frac{3}{4}$
- (C)  $\frac{4}{3}$
- (D)  $\frac{5}{2}$

**Answer:**





21. **Polynomial Remainder Theorem** (10 points)

For a polynomial  $p(x)$ , the value of  $p(3)$  is  $-2$ . Which of the following must be true about  $p(x)$ ?

- (A)  $x - 5$  is a factor of  $p(x)$ .
- (B)  $x - 2$  is a factor of  $p(x)$ .
- (C)  $x + 2$  is a factor of  $p(x)$ .
- (D) The remainder when  $p(x)$  is divided by  $x - 3$  is  $-2$ .

**Answer:**

